

Simple Linear Regression, Inference

CSU Chico, Math 314

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outline

Recap

Estimating β_0, β_1

R^2

Extrapolation

Take Away

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Recap, linear regression

Linear regression is a method to fit a line through a scatter plot of data in a “best” sense. Often, interest lies in the relationship between the explanatory and the response variable.

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Inference, β_0, β_1

Linear regression estimates the population parameters β_0 (intercept) and β_1 (slope), just like every other parameter we have estimated. As such, the estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ of these parameters have their own sampling distributions.

Inference, β_0, β_1

It turns out that the estimators $\hat{\beta}$ are also approximately normally distributed when the sample size is sufficiently large,

$$\frac{\hat{\beta} - \beta}{\sigma_{\hat{\beta}}} \sim N(0, 1).$$

Once again, we get this result through **standardization** (estimator minus mean over standard deviation).

Inference, hypothesis tests

Hypothesis testing naturally follows. The most common hypothesis test for linear regression parameters is

$$H_0 : \beta = 0$$

$$H_A : \beta \neq 0.$$

with $\alpha = 0.05$.

Inference, a note

A small but often missed point: hypothesis testing is meaningless if the model is wrong.

George E. P. Box

Essentially, all models are wrong, but some are useful.

Standard Output

The hypothesis test above has a natural and informative interpretation in most contexts.

```
url <- "https://raw.githubusercontent.com/roualdes/  
data/master/elmhurst.csv"  
elmhurst <- read.csv(url)  
elmReg <- lm(gift_aid ~ family_income, data=elmhurst)  
# summary(elmReg) # RStudio
```

Standard Output

Be sure to understand and be able to find

- ▶ standard errors
- ▶ t values
- ▶ p-values
- ▶ adjusted R^2

Inference, confidence intervals

If we can do hypothesis testing, we can do confidence intervals.
The function `confint` in R is extremely helpful.

```
# use lm fitted model, elmReg from above
```

```
confint(elmReg) # default is 95%
```

```
##                2.5 %          97.5 %
```

```
## (Intercept)    21.72269421  26.91596380
```

```
## family_income -0.06480555 -0.02133775
```

```
confint(elmReg, level=0.99) # can specify confidence
```

```
##                0.5 %          99.5 %
```

```
## (Intercept)    20.85539590  27.78326212
```

```
## family_income -0.07206486 -0.01407844
```

Testing with the finches data

Try the above steps with the [finches](#) data, predicting winglength using middletoelength as in the main lecture notes. What are all your relevant values from your standard output?

```
finches <- read.csv("https://raw.githubusercontent.com/njlytal/MATH314/main/finches.csv")
```

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R^2

It is common to use the square of the (Pearson) correlation to explain the strength of a linear fit.

R^2

The R^2 of a linear model describes the amount of variation in the response variable y that is explained by the least squares line on the explanatory variable x .

R^2 , example

Using the data frame `elmhurst`, the correlation between gift aid and family income is $R = -0.4986$. Thus, $R^2 = 0.2486$. We say 24.86% of the variation in `gift_aid` is explained by the least squares line on `family_income`.

Adjusted R^2

It is almost¹ always better to use the “Adjusted R-squared” value that R gives you in all linear regression output. The problem with the R^2 above is that the more explanatory variables you use the higher this value is – this doesn't always conform to reality.

¹Except when the adjusted R^2 is negative.

Adjusted R^2

adjusted R^2

The adjusted R^2 of a linear model describes the amount of variation in the response variable y that is explained by the least squares line on the explanatory variable x , after taking into account the number of explanatory variables.

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Extrapolation

Extrapolation is in general dangerous. Sometimes it works, but not often so watch out.

extrapolation

Applying a model to values outside of the range of the original data is called extrapolation.

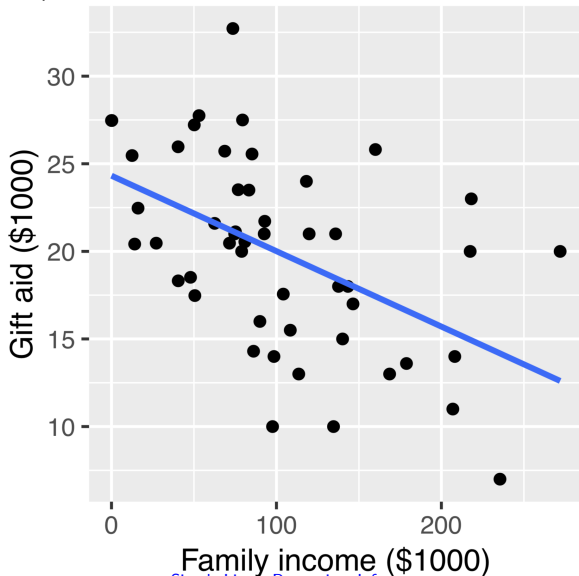
Extrapolation, example

A vehicle from a particular manufacturer that weighs 1800 lbs. has an mpg of 34, while another vehicle from the same manufacturer that weighs 2400 lbs. has an mpg of 22. At the current rate, what would be the mileage of an even larger vehicle that weighs 3600 lbs.?

What is the issue here?

Extrapolation, example

How much gift aid would a student expect to receive if their family income was \$1 million?



Extrapolation, example

How much would gift aid would a student expect to receive if their family income² was \$1 million? Using our least squares line,

$$\widehat{aid} = 24.32 + -0.04 \times family_income$$

we'd estimate -18.75 thousand dollars.

```
?predict.lm
```

²Don't forget family income is in units of \$1000.

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Take away

- ▶ Hypothesis testing and confidence intervals live on
- ▶ most linear regression software output defaults to two-sided alternatives
- ▶ Too many people rely on R^2 , use adjusted R^2 instead.
- ▶ Extrapolation is dangerous – be careful.

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References

references I

David M Diez, Christopher D Barr, and Mine Cetinkaya-Rundel. *OpenIntro Statistics*. CreateSpace independent publishing platform, third edition, 2015.

Hadley Wickham. *ggplot2: elegant graphics for data analysis*. Springer Science and Business Media, 2009.

Adapted from notes by Dr. Roualdes.